

Daniele Zago

Padova, Italy |  [dedzago.github.io](https://github.com/dedzago) |  [DedZago](#) |  zagodaniele.9@gmail.com |  [0000-0003-0778-7099](#)

Date of birth: 9 May 1996

Research-oriented data scientist and optimization engineer bridging **statistics**, **machine learning**, and **mathematical optimization**. I design and implement end-to-end solutions spanning stochastic optimization algorithms, ML models, and production software on enterprise stacks, with a track record in both academic research and industrial deployment.

WORK EXPERIENCE

Algorithm Engineer Sep 2026 – Present
Timefold Ghent, Belgium

TECHNOLOGIES: **Java**

Data Scientist Oct 2024 – Aug 2026
Optit S.r.l. Bologna, Italy

TECHNOLOGIES: **Python, Java, Hexaly (LocalSolver), Xpress Mosel, Gradle, Docker, scikit-learn, MLFlow**

- Research and development of solutions to the **periodic vehicle routing problem** with clustered routes for multi-day strategic fleet management, including front-end development for interactive visualization and exploration
- **MILP models** for customer delivery with cross-dock facilities, and stock balancing optimization with forecasted demand
- Developed **forecasting and anomaly detection** ML-based pipelines for power plant energy consumption

Statistical consultant Jul 2023 – Oct 2023
Expin S.r.l. Padua, Italy

TECHNOLOGIES: **R, C++**

Developed a sequential monitoring system for anomaly detection in vibrating string strain gauges, and a novel algorithm for optimal alarm threshold selection using stochastic optimization techniques.

Teaching assistant Oct 2022 – Dec 2022
Department of Developmental Psychology and Socialisation, University of Padua Padua, Italy

Lectures on factor models and analysis of questionnaires, laboratory sessions on programming.

Academic tutor Sep 2017 – Jan 2019
Department of Statistical Sciences, University of Padua Padua, Italy

Provided calculus lessons and interactive workshops for undergraduate students, in preparation for the exams.

EDUCATION

University of Padua Padua, Italy
Ph.D. in Statistical Sciences 2021 – 2024

- Advisor: prof. Giovanna Capizzi; Co-advisor: prof. Peihua Qiu
- Research topic(s): **online outlier detection** and **stochastic optimization**

University of Florida Gainesville, FL, USA
Visiting research scholar, supervisor: Prof. Peihua Qiu Jan 2023 – Dec 2023

INFN Bertinoro, Italy
Thirteenth INFN International School on Efficient Scientific Computing Oct 2022

- Efficient C++ programming, GPU programming with CUDA

University of Padua Padua, Italy
M.Sc. in Statistical Sciences 2019 – 2021

- Final grade: **110/110 cum Laude**, GPA: **29.5/30**
- Thesis topic(s): **Bayesian nonparametric mixture models**

University of Perugia Perugia, Italy
Summer school in Mathematics (functional analysis and mathematical statistics) Jul 2020

- Final grade: **110/110 cum Laude**, GPA: **29.2/30**
- Thesis topic(s): **applied Bayesian modelling**

SKILLS

EXPERTISE	Mathematical and stochastic optimization, constrained optimization, statistical process monitoring, machine learning, anomaly detection, statistical modelling
PROGRAMMING	R, Julia, Python, C++, Rust, Java, TypeScript, SQL, Stan, bash, SAS
TOOLS	git, MLFlow, Hexaly, Gradle, Docker, Microsoft Office
LANGUAGES	Italian (native), English (fluent, C2), German (moderate), Spanish (moderate)

PUBLICATIONS

Journal articles

Zago, D. (2025). StatisticalProcessMonitoring.Jl: A General Framework for Statistical Process Monitoring in Julia. *Journal of Statistical Software* 113, 1–45. doi: 10.18637/jss.v113.i07 (*open-source Julia package*)

Zago, D., Capizzi, G., and Qiu, P. (2025). An Improved Bisection-Type Algorithm for Control Chart Calibration. *Statistics and Computing* 35, 81. doi: 10.1007/s11222-025-10609-7

Zago, D., Tian, Z., Capizzi, G., and Qiu, P. (2025). A General Framework for Monitoring Mixed Data. *Journal of Quality Technology* 57, 282–296. doi: 10.1080/00224065.2025.2512164

Zago, D., Capizzi, G., and Qiu, P. (2024). Optimal Constrained Design of Control Charts Using Stochastic Approximations. *Journal of Quality Technology* 56, 257–275. doi: 10.1080/00224065.2024.2323585

Zago, D., and Capizzi, G. (2024). Alternative Parameter Learning Schemes for Monitoring Process Stability. *Quality Engineering* 36, 560–574. doi: 10.1080/08982112.2023.2253891

AWARDS

2025	ENBIS Knowledge Fund, ENBIS 2025 conference	<i>Piraeus, Greece</i>
2022	Young Travel Award, ISBA 2022 conference	<i>Montréal, Canada</i>
2018	Mille e una Lode Award 2018 (<i>top 3% of students</i>)	<i>University of Padua</i>
2017	Mille e una Lode Award 2017 (<i>top 3% of students</i>)	<i>University of Padua</i>

CONFERENCE PRESENTATIONS

Nov 2025	Invited seminar. <i>University of Padua</i>	<i>Padua, Italy</i>
	Efficient algorithms for control limit calibration	
Sep 2025	Invited talk. <i>ENBIS-25 Conference</i>	<i>Piraeus, Greece</i>
	Optimal constrained design of control charts using stochastic approximations	
Oct 2023	Invited talk. <i>2023 INFORMS Annual Meeting</i>	<i>Phoenix, AZ, USA</i>
	Optimal constrained design of control charts using stochastic approximations	
Sep 2022	Poster presentation. <i>Statistical methods and models for complex data</i>	<i>Padova, Italy</i>
	Profile monitoring based on adaptive parameter learning	
Jun 2022	Poster presentation. <i>2022 ISBA World meeting</i>	<i>Montréal, Canada</i>
	Bayesian nonparametric multiscale mixture models via Hilbert-curve partitioning	